

Slopes of Parallel and Perpendicular Lines

Using slopes to test whether segments are parallel or perpendicular, and to prove claims about figures on the coordinate plane

Directions

- Slope is the change in y over the change in x between two points on the line.
- Two non-vertical lines are **parallel** exactly when their slopes are equal, and **perpendicular** exactly when their slopes are opposite reciprocals — that is, when the product of the slopes is -1 .
- Show the slope calculation for every segment you compare. A conclusion with no slopes behind it earns no credit.
- Leave fractional slopes as fractions in lowest terms.

1. Find the slope of $\overline{AB}$ for $A(2, 1)$ and $B(6, 9)$. Then state the slope of any line parallel to $\overline{AB}$.

Answer: _____

2. Find the slope of $\overline{CD}$ for $C(-3, 5)$ and $D(1, -3)$. Then state the slope of any line perpendicular to $\overline{CD}$.

Answer: _____

3. $\overline{PQ}$ has endpoints $P(0, -2)$ and $Q(4, 4)$; $\overline{RS}$ has endpoints $R(-1, 3)$ and $S(3, 9)$. Are the two segments parallel? Find both slopes and justify your answer.

Answer: _____

4. $\overline{EF}$ has endpoints $E(1, 2)$ and $F(5, 4)$; $\overline{GH}$ has endpoints $G(0, 6)$ and $H(2, 2)$. Are the two segments perpendicular? Find both slopes and justify your answer.

Answer: _____

5. $\overline{JK}$ has endpoints $J(4, -1)$ and $K(4, 7)$. Find the slope of $\overline{JK}$. Then state the slope of any line perpendicular to $\overline{JK}$ and explain why the opposite-reciprocal rule has to be read carefully here.

Answer: _____

6. Quadrilateral $ABCD$ has vertices $A(-2, 1)$, $B(2, 3)$, $C(4, 7)$ and $D(0, 5)$. Find the slopes of all four sides and use them to show that both pairs of opposite sides are parallel.

Answer: _____

7. Triangle PQR has vertices $P(-1, -1)$, $Q(3, 1)$ and $R(1, 5)$. Use slopes to show that the angle at Q is a right angle, then find the area of triangle PQR in square units.

Answer: _____

8. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 2)$, $Y(8, 8)$ and $Z(2, 6)$.

(a) Find the slopes of $\overline{WX}$ and $\overline{XY}$ and decide whether $WXYZ$ is a rectangle.

(b) Find the midpoint of each diagonal and state what those midpoints show about the figure.

Answer: _____

9. Quadrilateral $EFGH$ has vertices $E(1, 2)$, $F(4, 5)$, $G(7, 2)$ and $H(4, -1)$. Find the slopes of $\overline{EF}$ and $\overline{FG}$, then find the lengths of those two sides to the nearest hundredth. Explain what the four results together prove about $EFGH$.

Answer: _____

10. Triangle TUV has vertices $T(0, 0)$, $U(6, 2)$ and $V(2, 8)$. Let M be the midpoint of $\overline{TV}$ and let N be the midpoint of $\overline{UV}$.

(a) Find the coordinates of M and N .

(b) Find the slope of $\overline{MN}$ and the slope of $\overline{TU}$.

(c) Write a short coordinate proof that $\overline{MN}$ is parallel to $\overline{TU}$.

Answer: _____